

EPPS6323 Knowledge Mining

Assignment 2 – Brush up R and Quarto

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EPPS: 6323

1. Lab01, Lab02 in R (Download the .qmd files in Teams)
 - a. Render and upload your files to GitHub website
 - b. Be sure to upload associated folders (e.g. Lab01.html, Lab01_files, Lab02.html, Lab02_files)
2. Review Chapters 3 to 7 in [R4DS](#), Section 3.5 in [Data programming](#). Run an exploratory data analysis with R using the TEDS2016 dataset (sample codes as follows):

```
library(haven) TEDS_2016
<-
read_stata("https://github.com/datageneration/home/blob/master/DataProgramming
/data/TEDS_2016.dta?raw=true")
```

What problems do you encounter when working with the dataset?

Ans:

- Had to copy the .dta file to local folder, have haven library installed and specify the correct folder path and load the dataset to open the .dta file
- Missing values were found in the dataset, had to handle them.
- Had to identify whether the different variables mentioned comes under categorical or numeric variables and correspondingly assign statistical methods to study relationships.

3. How to deal with missing values?

Ans:

- **Checked for missing values for all columns**, `colSums(is.na(TEDS_2016))`,
- **Checked for these particular columns and it was found that edu has 10 missing values** `colSums(is.na(TEDS_2016[, c("Tondur", "female", "DPP", "age", "income", "edu", "Taiwanese", "Econ_worse"))])`.
- **The 10 missing values were omitted from the column edu using filter that comes under the dplyr package:** `TEDS_2016 <- TEDS_2016 %>% filter(!is.na(edu))`

4. (Next step) Explore the relationship between Tondur and other variables including female, DPP, age, income, edu, Taiwanese and Econ_worse. What methods would you use?

Ans:

Tondur looks a categorical variable as it is seen to represent different views on the Taiwan independence–unification question. Here, when opened in R studio Tondur appears to be measured on a 1 to 9 scale, where:

- 1 for favoring unification with China
- 9 for support to independence
- Middle values like 4, 5 representing support for the status quo

Methods Used

Used mainly three statistical methods here to find relationship between Tondu and the rest of the variables.

- **Spearman's Rank Correlation**, a non-parametric test is used to measure relationships between Tondu and perceivably numeric predictors including **age, income, edu and Econ_worse**. A **positive correlation** associates with a more pro-independence stance, whereas a **negative correlation** suggests a tendency towards unification.
- **Chi-Square tests** examine Tondu's association to **categorical variables** namely **gender (female), party preference (DPP) and identity (Taiwanese)**, to help identify political leaning. Associations are significant for **p-value below 0.05**.
- **Ordinal Logistic Regression** stresses multiple variables and how it affects the overall result giving an idea on which factors increase or decrease the likelihood of backing independence. Values **greater than 1 suggests a preference of pro-independence views** and **less than 1 for supporting unification**.

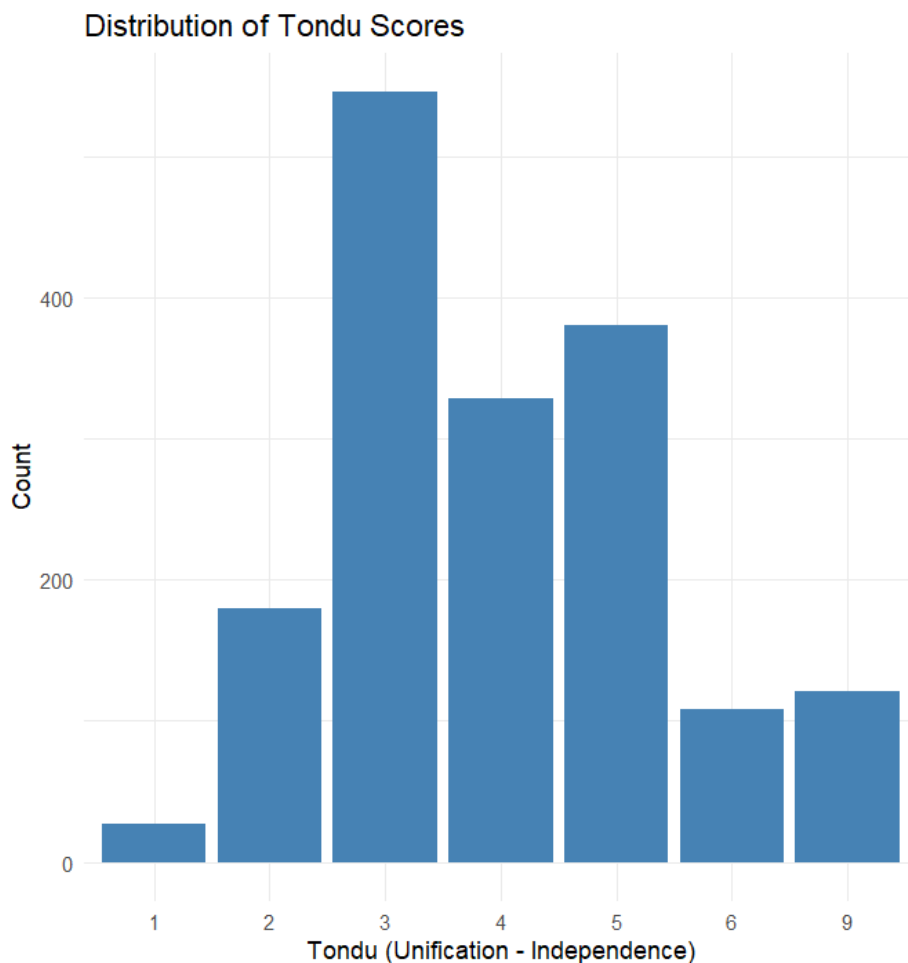


Chart shows the **distribution of Tondu scores**, which represent people's stance on the **Independence–Unification issue** in Taiwan.

1-representing strong Unification with china

9-representing remaining an independent country

3-5 represents status quo.

Politically what it means :

Majority of respondents prefer maintaining the status quo (3-5 range).

Extreme positions (1 = Strong Unification, 9 = Strong Independence) are uncommon.

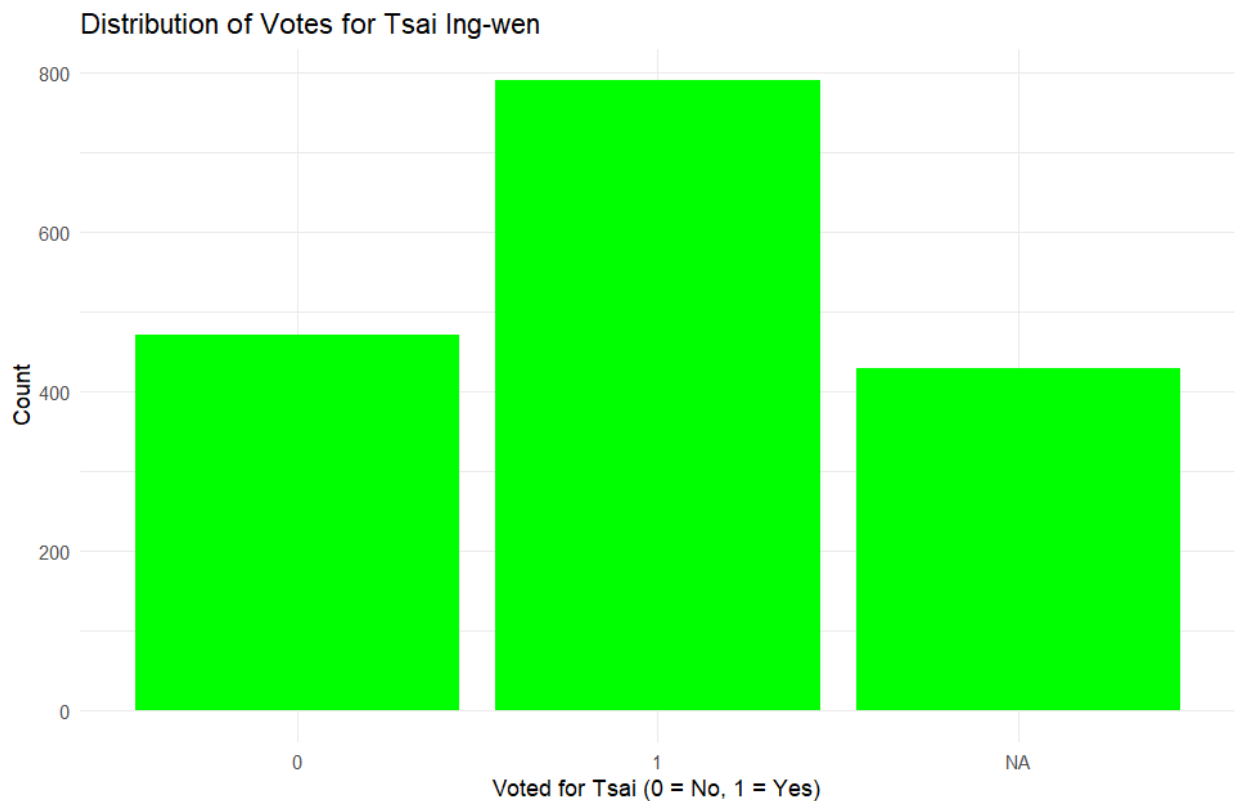
More people lean towards pro-independence (5) than pro-unification (1-2).

This aligns with Taiwan's political reality, where most people prefer stability rather than an immediate change toward independence or unification.

5. (Next step) How about the `votetsai` variable (vote for DPP candidate Tsai Ing-wen)?

No. of people who voted 790

Percentage of vote 62%



```

63
64 #Distribution of votesai votes
65 table(TEDS_2016$votesai) # Frequency count of people who voted for Tsai
66 prop.table(table(TEDS_2016$votesai)) # Proportion of votes
67
68

```

67:1 (Top Level) R Script

Console Terminal Background Jobs

R 4.4.2 · C:/Users/Public/Documents/Ann_Quarto_Webpage/

```

> # bar plot of Tondur distribution
> ggplot(TEDS_2016, aes(x = Tondur)) +
+   geom_bar(fill = "steelblue") +
+   theme_minimal() +
+   labs(title = "Distribution of Tondur Scores", x = "Tondur (Unification - Independenc
e)", y = "Count")
> table(TEDS_2016$votesai) # Frequency count of people who voted for Tsai

  0    1
471 790
> prop.table(table(TEDS_2016$votesai)) # Proportion of votes

      0      1
0.3735131 0.6264869
> |

```

6. (Next step) Generate frequency table and barchart of the Tondur variable. Assign labels to the variable using the following:

```

> #generate a table of counts:
> table(TEDS_2016$Tondur) # Frequency counts of each category

      Unification now      Status quo, unif. in future
              27                      180
      Status quo, decide later      Status quo forever
              546                      328
      Status quo, indep. in future      Independence now
              380                      108
              No response
              121
> prop.table(table(TEDS_2016$Tondur)) # Proportion of each category

      Unification now      Status quo, unif. in future
      0.01597633          0.10650888
      Status quo, decide later      Status quo forever
      0.32307692          0.19408284
      Status quo, indep. in future      Independence now
      0.22485207          0.06390533
              No response
      0.07159763

```



```
TEDS_2016$Tondu<-as.numeric(TEDS_2016$Tondu,labels=c("Unification now", "Status quo, unif. in future", "Status quo, decide later", "Status quo forever", "Status quo, indep. in future", "Independence now", "No response"))
```

What it all concludes to:

- Most **Taiwanese people prefer maintaining the status quo** rather than immediate changes.
- More people **favor future independence (22.49%) than future unification (10.65%)**.
- **Extreme positions (immediate unification or independence) are unpopular among the Taiwanese larger population.**